

String Data Type

A group of characters enclosed within single or double or triple quotes is called a string. So we can say the string is a sequential collection of characters.

1. Collection of character's enclosed within single or double or triple quotes is called a string.
2. Duplicates are allowed.
3. Ordered/Sequential Collection.
4. Indexing Supported.
5. Slicing Supported.
6. Duplicate characters are allowed.
7. Immutable in nature.

```
s1 = "Welcome to 'python' learning"
s2 = 'Welcome to "python" learning'
s3 = """Welcome "to" 'python' learning"""
print(s1)
print(s2)
print(s3)
# Output
Welcome to 'python' learning
Welcome to "python" learning
Welcome "to" 'python' learning
```

Note: Can we use single quote inside single quote or double quote inside double quote or triple quote inside triple quote.

```
s1 = "Welcome to \"python\" learning"
s2 = 'Welcome to \'python\' learning'
s3 = """Welcome \'\'to\'\' python learning"""
print(s1)
print(s2)
print(s3)
# Output
Welcome to "python" learning
Welcome to 'python' learning
Welcome \'\'to\'\' python learning
```

Python In-built functions for string:-

Step	Syntax	Explanation
1	max()	To check which object/character may have maximum ASCII value in given string
2	min()	To check which object/character may have minimum ASCII value in given string.
3	type()	To check data-type of an object.
4	id()	To check memory address of an object.
5	print()	To take output in a terminal.
6	len()	To check how many objects/characters present in string.
7	str()	to create empty string or used to type casting.
8	ord()	Character to ASCII value.
9	chr()	ASCII value to character.

Example:

```
str1 = "Cybrom Technology"
```

```
print(max(str1))
```

```
print(min(str1))
```

```
print(len(str1))
```

```
print(type(str1))
```

```
print(id(str1))
```

```
# Output
```

```
y
```

```
17
```

```
<class 'str'>
```

```
135446907837040
```

```
str1 = "Cybrom Technology"
```

```
for i in str1:
```

```
    print(f' ASCII value of {i} = {ord(i)}')
```

```
# Output
```

```
ASCII value of C = 67
```

```
ASCII value of y = 121
```

```
ASCII value of b = 98
```

```
ASCII value of r = 114
```

```
ASCII value of o = 111
```

```
ASCII value of m = 109
```

```
ASCII value of = 32
```

```
ASCII value of T = 84
```

```
ASCII value of e = 101
ASCII value of c = 99
ASCII value of h = 104
ASCII value of n = 110
ASCII value of o = 111
ASCII value of l = 108
ASCII value of o = 111
ASCII value of g = 103
ASCII value of y = 121
```

```
for i in range(65,75):
    print(f'Charactor of given value of {i} = {chr(i)}')
```

Output

```
Charactor of given value of 65 = A
Charactor of given value of 66 = B
Charactor of given value of 67 = C
Charactor of given value of 68 = D
Charactor of given value of 69 = E
Charactor of given value of 70 = F
Charactor of given value of 71 = G
Charactor of given value of 72 = H
Charactor of given value of 73 = I
Charactor of given value of 74 = J
```

String Methods

1. upper()
2. lower()
3. title()
4. capitalize()
5. swapcase()
6. find()
7. index()
8. count()
9. startswith()
10. endswith()
11. isalpha()
12. isdigit()
13. isalnum()
14. islower() & isupper()

15. replace()

16. split()

17. join()

18. ord()

19. chr()

upper():--

```
s = "python"
print(s.upper())
```

O/P:- PYTHON

lower():--

```
s = "PYTHON"
print(s.lower())
```

O/P:- python

title():--

```
s = "welcome to python"
print(s.title())
```

O/P:- Welcome To Python

capitalize():--

```
s = "python language"
print(s.capitalize())
```

O/P:- Python Language

swapcase():--

```
s = "PyThOn"
print(s.swapcase())
```

O/P:- pYtHoN

find():-- Returns first index, otherwise -1

```
s = "hello python"
print(s.find("python"))
```

O/P:- 6

index():-- Same as find() but gives error if not found

```
s = "hello python"
print(s.index("python"))
```

O/P:- 6

count():--

```
s = "python is easy and python is fun"
print(s.count("python"))
```

O/P:- 2

startswith():--

```
s = "python"
print(s.startswith("py"))
```

O/P:- True

endswith():--

```
s = "python"
print(s.endswith("on"))
```

O/P:- True

isalpha():--

```
s = "python"
print(s.isalpha())
```

O/P:- True

isdigit():--

```
s = "python"
print(s.isdigit())
```

O/P:- False

isalnum():--

```
s = "Python123"
print(s.isalnum())
```

O/P:- True

islower() & isupper():--

```
s = "python"
print(s.islower())
print(s.isupper())

# O/P:- True
#                False
```

replace():--

```
s = "I like Java"
print(s.replace("Java", "Python"))

# O/P:-- I like Python
```

split():--

```
s = "python is easy"
print(s.split())

# O/P:-- ['python', 'is', 'easy']
```

join():--

```
words = ['python', 'is', 'easy']
print(" ".join(words))

# O/P:-- python is easy
```

ord():-- char to ASCII

```
words = "A"
print(ord('A'))

# O/P:-- 65
```

chr():-- ASCII to char

```
ascii_val = 65
print(chr(ascii_val))

# O/P:-- A
```